

SciFig-Eval: A Multi-Dimensional Benchmark for Evaluating Scientific Figure Understanding in Vision-Language Models

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Abstract

We present SCIFIG-EVAL, a benchmark for evaluating vision-language models (VLMs) on scientific figure understanding across two complementary axes, open-ended description and targeted reasoning, each under standard and adversarial conditions. We evaluate 8 VLMs on 250 English scientific figures using checklist-based MQM scoring validated against human judgment (Spearman $\rho = 1.0$, $p < 0.001$), 6 image transforms, psychology-informed hallucination probes grounded in presupposition embedding and anchoring bias, and caption bias resistance testing. We introduce the Admittance-Resistance-Inductance (A-R-I) framework, which decomposes behavioral reliability into independently measurable dimensions through selective blur experiments. Our results reveal that quality and reliability rankings diverge at consequential points. GPT-5.2 ranks first on description quality (MQM 91.6) but admits visual limitations only 6% of the time, while Gemini 3.1 Pro admits 90% and leads on resistance (0.91). Phi-4 follows poisoned captions 95% of the time despite producing coherent descriptions. These divergences, invisible to quality-only benchmarks, demonstrate that current evaluation practices provide an incomplete picture of model readiness for scientific deployment.

1 Introduction

The model that writes the best scientific figure descriptions is also the one that most confidently fabricates information it cannot see. Vision-language models are rapidly entering scientific workflows, from literature review assistants that summarize chart findings to accessibility tools that narrate figures for visually impaired researchers. When a model misreads a trend line or invents a data point, the downstream consequences propagate silently: a fabricated value becomes a cited fact, a hallucinated comparison shapes a research conclusion.

The stakes demand not only that models describe figures accurately, but that they behave reliably when faced with ambiguity, missing information, or misleading context. Yet current evaluation practices assess only the first of these requirements, leaving the second entirely unmeasured.

Existing benchmarks for scientific figure understanding evaluate models exclusively through question answering on clean images. FigureQA (?) and PlotQA (?) test binary and numerical reasoning over synthetic or semi-synthetic charts. ChartQA (?) advances to human-authored questions on real-world figures, and ChartBench (?) broadens chart type coverage. These benchmarks measure whether a model produces a correct answer, but none ask whether a model admits when it cannot see, resists false premises embedded in questions, or distinguishes its own visual evidence from authoritative-sounding but incorrect textual context. A model can score highly on ChartQA while confidently fabricating information about chart elements that do not exist, because no existing benchmark tests for this failure mode.

We present SCIFIG-EVAL, a diagnostic evaluation framework that measures both description quality and behavioral reliability on scientific figures. The framework is organized around two evaluation axes (open-ended description and targeted reasoning) crossed with two conditions (standard and adversarial), yielding a 2×2 evaluation matrix that structures every experiment. Adversarial probes are grounded in cognitive science: *presupposition embedding* (?) tests whether definite articles trick models into accepting non-existent chart elements, *anchoring bias* (?) plants plausible but incorrect values, and *sycophantic agreement* (?) measures whether models echo false captions over their own visual perception. These probes feed into the A-R-I *behavioral framework*, which decomposes model behavior into three independently measurable dimensions: Admittance (does the model acknowl-

edge visual limitations?), Resistance (does it push back on false premises?), and Inductance (can it infer missing information from context?). We evaluate 8 models across 250 figures and over 11,000 evaluation instances (Figure ??).

Quality rankings diverge from reliability rankings at precisely the points that matter for deployment. GPT-5.2 leads on description quality with an MQM score of 91.6, but admits visual limitations only 6% of the time when asked about elements it cannot see, fabricating answers for the remaining 94%. Gemini 3.1 Pro, ranking second on quality at 90.2, admits uncertainty 90% of the time and achieves the highest resistance to hallucination probes (0.91). The divergence extends to caption dependency: Phi-4 follows poisoned captions over its own visual evidence 95% of the time ($R = 0.05$), while presupposition embedding, the technique borrowed from eyewitness testimony research, catches models that successfully resist explicit false values but fail against implicit assumptions about non-existent elements. A benchmark reporting only quality scores would rank GPT-5.2 and Gemini as near-equivalent, missing the fact that one fabricates 98% of the time while the other acknowledges uncertainty.

Our contributions are as follows:

1. A multi-dimensional evaluation framework for scientific figure understanding that pairs standard quality assessment with psychology-informed adversarial probes across two evaluation axes and two conditions (§3).
2. The A-R-I behavioral framework, which decomposes model behavior into Admittance, Resistance, and Inductance, providing independently measurable dimensions of reliability under uncertainty (§3.4).
3. An empirical demonstration that description quality and behavioral reliability are distinct dimensions of VLM competence, with model rankings diverging substantially across these dimensions (§4).
4. A full benchmark release including the annotated dataset, adversarial probes, evaluation code, and an interactive review dashboard.

2 Related Work

2.1 Chart and Figure Understanding Benchmarks

Chart understanding benchmarks have progressed from synthetic binary questions (?) to numerical reasoning over real-world plots (??) and complex visual reasoning across diverse chart types (?). ? targeted scientific graphs specifically, generating multi-turn QA from arXiv papers. A comprehensive comparison of these and additional benchmarks appears in Appendix ??.

These benchmarks share a common limitation. They evaluate VLMs exclusively through question answering on clean images, measuring whether a model produces a correct answer. None assess behavioral reliability, whether a model fabricates information it cannot see, echoes false premises, or acknowledges visual uncertainty. SCIFIG-EVAL addresses this gap by pairing standard quality evaluation with adversarial behavioral probes, measuring not only *what* a model says but *how* it behaves under uncertainty and deception.

2.2 VLM Hallucination and Reliability

Hallucination in vision-language models has received growing attention. ? introduced the CHAIR metric for detecting object hallucination in image captions. ? proposed POPE, converting hallucination detection into a polling-based binary classification task that systematically probes co-occurrence biases. ? designed HallusionBench with expert-crafted visual illusions and language traps to diagnose entangled hallucination modes, and ? introduced MMHal-Bench to penalize hallucination across eight question categories. ? provide a comprehensive survey categorizing hallucination sources in multimodal models.

These efforts focus predominantly on natural images, where hallucination means inventing objects absent from a scene. Scientific figures present distinct challenges, with structured layouts, quantitative data encoded in visual channels, and domain conventions that demand precise extraction rather than approximate recognition. Moreover, no existing benchmark tests whether a model *admits* when visual information is unavailable.

Our probes draw on cognitive science rather than treating hallucination as a monolithic category. *Presupposition embedding* (?) uses definite articles to make models accept non-existent chart elements. *Anchoring bias* (?) plants plausible but incorrect

values that models fail to challenge. The *cooperative principle* (?) exploits models’ assumption that questions are well-formed, making them answer rather than refuse. *Sycophantic agreement* (?) captures models’ tendency to echo provided captions over their own visual evidence.

2.3 Evaluation Methodology

SCIFIG-EVAL adapts the Multidimensional Quality Metrics (MQM) framework (?), originally developed for translation quality assessment and subsequently adopted as the standard for WMT human evaluation (?). MQM decomposes quality into fine-grained error categories with severity weightings, enabling diagnostic evaluation rather than holistic scoring. We extend MQM to chart descriptions by introducing *binding verification*, which checks not only whether described elements exist in the figure but whether stated relationships between elements (e.g., “the blue bar exceeds the red bar”) are grounded in the visual evidence.

For scalable evaluation, we follow the LLM-as-judge paradigm validated by ?, who demonstrated strong agreement between LLM judges and human preferences on MT-Bench. Our judges apply checklist-based scoring in which each figure is annotated with verifiable claims, and the judge evaluates whether the model’s description correctly addresses each claim. This structured approach mitigates the subjectivity of holistic ratings while enabling the scale required to evaluate eight models across multiple conditions.

Together, these methodological choices position SCIFIG-EVAL as the first framework that unifies description quality measurement with behavioral reliability assessment on scientific figures. Existing benchmarks measure quality *or* detect hallucinations; we measure both and find that model rankings diverge substantially across these dimensions (§??).

3 Evaluation Framework

We organize SCIFIG-EVAL around two complementary axes of evaluation: *what* a model communicates (open-ended description vs. targeted question answering) and *how* the model behaves under pressure (standard vs. adversarial conditions). This yields a 2×2 evaluation matrix (Table 1) that structures every experiment in the benchmark.

Under standard conditions, we measure description quality through checklist-based MQM scoring

	Standard	Adversarial
Description	Baseline MQM	Caption bias, passive admittance / inductance
Reasoning	Capability Qs	Resistance probes, active admittance / inductance

Table 1: The 2×2 evaluation matrix. Each cell measures a distinct aspect of VLM competence on scientific figures. Standard conditions assess quality; adversarial conditions expose behavioral reliability.

and reasoning ability through targeted capability questions (counting, computation, comparison, pattern analysis). Under adversarial conditions, we probe behavioral reliability by testing whether a model echoes false captions, accepts non-existent chart elements, or fabricates answers for information it cannot see. A key empirical observation motivating this design is that rankings under standard conditions diverge substantially from rankings under adversarial conditions, suggesting that quality and reliability are distinct dimensions of model competence.

3.1 Checklist-Based MQM Evaluation

We evaluate open-ended descriptions using a chart-type-specific variant of the Multidimensional Quality Metrics (MQM) framework. Each chart type has a dedicated checklist: 14 items for bar charts, 15 for line plots, and 11 for pie charts. Items cover structural identification (chart type, orientation, grouping), axis and legend accuracy, numerical correctness, and visual element description. Each item is classified as *Major* or *Minor* based on its impact on figure comprehension.

An LLM judge (GPT-4o) assesses each checklist item on two axes: *coverage* (complete, partial, or missing) and *correctness* (correct, partial, wrong, or n/a), then scans for violations of seven global constraints. A rule-based engine maps judge output to penalties across three dimensions (Accuracy, Completeness, and Clarity) with severity-dependent weights (full weight table in Appendix B). The final MQM score normalizes total penalties against a severity-adjusted maximum:

$$\text{MQM} = \max\left(0, 100 - \frac{\sum_i w_i}{D} \times 100\right) \quad (1)$$

where w_i is the weight of penalty i and D is the maximum possible penalty given the checklist composition.

Two mechanisms guard against scoring artifacts. *Binding verification* cross-checks element-level

value-label-colour correspondences, escalating any mismatch to a Major error. A *deduplication engine* collapses overlapping penalties from global constraints and checklist items. To validate automated scoring, two expert annotators independently ranked all eight models on a 30-figure subset. The automated MQM rankings achieve a Spearman correlation of $\rho = 1.0$ ($p < 0.001$, $n = 8$ models) with the human consensus ranking, confirming that the checklist-based approach preserves human judgment at scale.

3.2 Caption Bias Probes

Caption bias probes test whether a model trusts its own visual perception over misleading textual context. For each figure, we generate a modified caption containing exactly 2–3 embedded false claims, following a set of psychology-informed design principles. The *70/30 rule* ensures that roughly 70% of the modified caption is verifiably correct, establishing trust before introducing poisoned claims. False numerical values occupy an *anchoring sweet spot* of 20–40% deviation from actual values, close enough to be plausible yet clearly falsifiable by visual inspection (?). Claims target secondary details rather than dominant features (*peripheral misinformation*), and framing employs *loaded verbs* that subtly mischaracterize trends (e.g., “declined” vs. “stabilized”).

Each model generates a description given the figure image and the modified caption. An LLM judge then evaluates each false claim independently through a randomized A/B design: for each modification, the judge receives the claim and the ground-truth reality in randomized order (A or B) and determines which the model’s description aligns with, eliminating position bias. The *caption bias resistance score* is:

$$R_{\text{caption}} = \frac{n_{\text{image}}}{n_{\text{image}} + n_{\text{caption}}} \quad (2)$$

where n_{image} is the number of claims where the model followed visual evidence and n_{caption} is the number where it echoed the false caption.

3.3 Hallucination Resistance Probes

We design three probe types grounded in cognitive science to test whether models can resist fabricating information under adversarial questioning.

Inexist probes use *presupposition embedding* (?): definite articles (“the error bars,” “the regres-

sion line”) presuppose the existence of chart elements that are absent. The false element is embedded in a subordinate clause, directing attention to an analytical question *about* the element rather than *whether* it exists, exploiting co-occurrence priors for the chart type.

Contra probes exploit *anchoring bias* (?): a specific but incorrect numerical value (20–30% off) is embedded as a premise, and the model is asked to build upon it (e.g., compute a comparison). The model must first detect the false anchor, then correct it, which is substantially harder than reading a value directly.

Unanswerable probes leverage *conversational implicature* (?): domain-appropriate questions (requesting p -values, sample sizes, or projections beyond the plotted range) sound like legitimate analytical follow-ups but cannot be answered from the chart alone. Refusal is costly because the question sounds helpful and reasonable.

Each probe response is scored on a three-point rubric: 1.0 (correctly resists), 0.5 (hedges but partially complies), and 0.0 (fully accepts the false premise or fabricates an answer). The aggregate *resistance score* averages across all probes per model.

3.4 The A-R-I Behavioral Framework

The evaluation components described above feed into a unified behavioral theory that decomposes model behavior into three independently measurable dimensions, which we term the *A-R-I framework*.

Admittance measures whether a model acknowledges visual limitations rather than fabricating confidently. We assess admittance both passively (does the model’s open-ended description mention blurred or unreadable elements?) and actively (when asked a targeted question about a selectively blurred element, does the model admit uncertainty?). Active admittance evaluation uses a judge that independently scores two binary dimensions: whether the model *admits* uncertainty and whether it *fabricates* a specific answer. A model can do both simultaneously (e.g., “the label is obscured but appears to be X”), and this decomposition captures important behavioral nuance.

Resistance measures a model’s ability to push back against misleading inputs. This dimension aggregates the hallucination resistance probes (§3.3) and caption bias probes (§3.2), capturing whether

the model prioritizes its own visual evidence over authoritative-sounding but false textual context.

Inductance measures whether a model can infer missing information from surrounding context. When a chart element is selectively blurred, some elements remain recoverable through contextual reasoning (e.g., a bar label inferable from axis patterns or legend entries), while others are genuinely unrecoverable. We assess inductance through the same selective blur methodology used for admittance, but focus on the *correctness* of fabricated answers. When a model fabricates an answer for an inferable element, correctness validates whether genuine reasoning occurred or whether the output was a lucky guess. As we show in §4, this distinction is empirically validated by a large gap in fabrication correctness between inferable and non-inferable elements, confirming that inductance captures real reasoning rather than noise.

Independence of dimensions. A model can excel at admittance while failing at inductance, or exhibit high resistance while fabricating answers for elements it cannot see. As we demonstrate in §4, models that rank highest on description quality do not necessarily rank highest on behavioral reliability, and the gap can be substantial. Implementation details for selective blur appear in Appendix ??.

4 Experiments and Results

Model rankings under behavioral evaluation diverge substantially from quality rankings, revealing that description quality and behavioral reliability are distinct dimensions of VLM competence. We evaluated eight vision-language models across 250 scientific figures and over 14 experimental conditions, yielding more than 11,000 individual evaluation instances.

4.1 Models

We select eight models spanning commercial and open-weight families, dense and mixture-of-experts (MoE) architectures, and a range of effective parameter counts. **Commercial models:** GPT-5.2 (Azure, closed-weight), Gemini 3.1 Pro (closed-weight), and Phi-4-Multimodal (small, closed-weight). **Open-weight MoE models:** Llama 4 Maverick (17B params, 128 experts), Qwen3-VL-235B-A22B (235B total, 22B active), and Qwen3-VL-30B-A3B (30B total, 3B active). **Open-weight dense models:** Qwen3-VL-8B and Gemma3-27B-IT. This selection enables controlled comparisons

along three axes: commercial vs. open-weight deployment, MoE vs. dense architecture at similar active parameter counts (Qwen-235B at 22B active vs. Gemma at 27B dense), and scaling within a single family (the three Qwen variants from 3B to 22B active parameters). All models received identical prompts and were evaluated at temperature 0 with GPT-4o as the automated judge.

4.2 Description Quality

GPT-5.2 and Gemini produce the highest-quality scientific figure descriptions, but the gap between them is modest and a clear tier structure emerges across all eight models (Table 2). GPT-5.2 leads with a baseline MQM of 91.6 (95% CI: 90.4–92.8), followed by Gemini at 90.2 (88.9–91.4). The 1.4-point difference is statistically significant ($p < 0.01$) but practically small. A second tier clusters Llama 4, Qwen-235B, and Qwen-8B between 78.9 and 81.4, with overlapping confidence intervals. Qwen-30B (74.4) and Gemma (69.1) form a third tier, while Phi-4 trails at 62.2 (59.3–64.9), nearly 30 points behind the leaders. Accuracy penalties dominate across all models; completeness penalties become the primary differentiator for weaker models, with Phi-4 accumulating 8.8 completeness penalty points compared to GPT-5.2’s 0.7.

Transform robustness. Rotation is the most damaging transform, causing an average MQM drop of 19.4 points across models, while noise has negligible impact (Table 2, Noise column). Low contrast reduces scores by a moderate 4–7 points for top-tier models. The most striking result involves in-paper-blur, where the figure is embedded in a PDF page and the surrounding page content is blurred. This condition collapses all models below 31 MQM points, with Gemini dropping from 90.2 to 8.4 and GPT-5.2 from 91.6 to 13.3. We attribute this catastrophic failure to the models attempting to describe the blurred page context rather than focusing on the embedded figure, a finding with practical implications for document-level figure understanding pipelines.

Selective blur and caption bias MQM. Selectively blurring unrecoverable elements (admittance blur) reduces MQM by roughly 8–10 points for top models, while blurring inferable elements (inductance blur) produces smaller drops of 1–3 points (Table 2, AdmB and IndB columns). This difference validates the admittance-inductance distinc-

Model	Base	Orig.	Noise	Rot.	LowC	InPap	IPBlr	CapB	AdmB	IndB
GPT-5.2	91.6 ^{92.8} _{90.4}	89.6 ^{91.8} _{87.1}	91.3 ^{93.3} _{89.1}	70.2 ^{73.9} _{66.5}	87.0 ^{89.6} _{84.1}	77.8 ^{81.3} _{74.2}	13.3 ^{17.8} _{9.2}	91.3 ^{93.3} _{89.2}	81.7 ^{85.6} _{77.4}	88.9 ^{93.2} _{84.0}
Gemini	90.2 ^{91.4} _{88.9}	89.3 ^{91.3} _{87.2}	90.2 ^{92.1} _{88.2}	72.0 ^{75.3} _{68.6}	85.1 ^{87.4} _{82.7}	83.8 ^{86.3} _{81.1}	8.4 ^{12.1} _{4.9}	91.3 ^{93.2} _{89.4}	84.6 ^{87.5} _{81.5}	86.8 ^{90.4} _{82.7}
Llama 4	81.4 ^{83.3} _{79.5}	82.0 ^{84.9} _{78.9}	81.0 ^{84.0} _{77.9}	60.6 ^{64.7} _{56.3}	75.9 ^{78.8} _{72.7}	63.0 ^{66.9} _{58.9}	30.9 ^{35.2} _{26.7}	84.8 ^{87.5} _{81.7}	70.5 ^{75.1} _{65.7}	78.9 ^{83.5} _{74.2}
Qwen-235B	80.8 ^{82.7} _{78.9}	82.6 ^{85.3} _{79.8}	82.1 ^{84.7} _{79.6}	65.3 ^{68.6} _{62.0}	77.8 ^{80.6} _{74.7}	77.2 ^{80.7} _{73.6}	22.9 ^{27.7} _{18.3}	83.4 ^{85.7} _{80.9}	71.0 ^{76.7} _{65.2}	79.0 ^{83.1} _{74.5}
Qwen-8B	78.9 ^{81.1} _{76.8}	78.5 ^{81.8} _{75.0}	80.1 ^{83.2} _{76.8}	61.9 ^{65.4} _{58.2}	76.0 ^{79.5} _{72.4}	72.9 ^{76.9} _{68.9}	25.6 ^{30.4} _{21.3}	83.0 ^{85.8} _{80.1}	71.5 ^{76.8} _{66.1}	74.8 ^{80.2} _{68.9}
Qwen-30B	74.4 ^{76.7} _{72.1}	77.3 ^{80.7} _{73.6}	74.3 ^{78.0} _{70.6}	58.2 ^{62.1} _{54.1}	75.7 ^{78.9} _{72.5}	68.3 ^{72.5} _{63.7}	24.3 ^{28.7} _{19.9}	78.3 ^{81.6} _{74.7}	67.5 ^{73.1} _{61.5}	74.7 ^{80.0} _{69.1}
Gemma	69.1 ^{71.6} _{66.6}	60.9 ^{65.1} _{56.7}	62.2 ^{66.5} _{58.0}	49.8 ^{54.1} _{45.7}	61.5 ^{65.3} _{57.8}	35.4 ^{39.6} _{31.5}	25.3 ^{29.1} _{21.6}	70.9 ^{75.8} _{65.6}	58.3 ^{63.0} _{53.7}	59.6 ^{66.6} _{52.4}
Phi-4	62.2 ^{64.9} _{59.3}	59.5 ^{64.6} _{54.2}	61.3 ^{65.3} _{57.1}	43.7 ^{48.0} _{39.4}	64.7 ^{68.9} _{60.5}	31.3 ^{36.3} _{26.4}	17.7 ^{21.1} _{14.4}	61.7 ^{67.1} _{56.1}	56.6 ^{62.3} _{50.3}	59.4 ^{66.1} _{52.6}

Table 2: Description quality (MQM scores, 0–100) across conditions. Base = baseline with caption (250 figures). Transforms evaluated on 100-figure subset. Values show mean with 95% bootstrap CI (subscript/superscript). Higher is better.

Model	Resistance			Cap.	Active		Passive	
	Inex.	Cont.	Unans.	Bias R	Adm.	Ind. ^C	Adm.	Ind. ^C
GPT-5.2	0.77 ^{0.82} _{0.72}	0.75 ^{0.80} _{0.69}	0.92 ^{0.95} _{0.89}	0.89 ^{0.93} _{0.85}	6 ¹⁴ ₀	74 ⁸⁷ ₆₂	26 ³⁸ ₁₄	68 ⁸² ₅₅
Gemini	0.88 ^{0.92} _{0.84}	0.91 ^{0.94} _{0.87}	0.95 ^{0.97} _{0.92}	0.89 ^{0.93} _{0.85}	90 ⁹⁸ ₈₂	80 ⁹³ ₆₈	74 ⁸² ₆₀	66 ⁸⁰ ₅₁
Llama 4	0.63 ^{0.69} _{0.57}	0.76 ^{0.81} _{0.71}	0.94 ^{0.97} _{0.91}	0.74 ^{0.81} _{0.67}	22 ³⁴ ₁₂	43 ⁵⁷ ₃₀	4 ¹⁰ ₀	58 ⁷¹ ₄₂
Qwen-235B	0.67 ^{0.73} _{0.62}	0.64 ^{0.70} _{0.58}	0.94 ^{0.97} _{0.91}	0.54 ^{0.62} _{0.46}	12 ²² ₄	35 ⁵⁰ ₂₃	14 ²⁴ ₅	56 ⁶⁹ ₄₂
Qwen-8B	0.40 ^{0.46} _{0.34}	0.44 ^{0.50} _{0.38}	0.88 ^{0.92} _{0.84}	0.43 ^{0.51} _{0.34}	6 ¹⁴ ₀	30 ⁴³ ₁₇	6 ¹⁴ ₀	62 ⁷⁶ ₄₇
Qwen-30B	0.23 ^{0.29} _{0.18}	0.37 ^{0.43} _{0.31}	0.73 ^{0.79} _{0.68}	0.30 ^{0.38} _{0.22}	6 ¹⁴ ₀	29 ⁴² ₁₇	2 ⁶ ₀	61 ⁷⁵ ₄₈
Gemma	0.17 ^{0.22} _{0.13}	0.24 ^{0.30} _{0.19}	0.93 ^{0.96} _{0.90}	0.38 ^{0.46} _{0.30}	2 ⁶ ₀	21 ³³ ₁₀	2 ⁶ ₀	44 ⁶⁰ ₃₁
Phi-4	0.04 ^{0.06} _{0.02}	0.04 ^{0.06} _{0.02}	0.56 ^{0.62} _{0.49}	0.05 ^{0.09} _{0.02}	2 ⁶ ₀	21 ³⁴ ₁₁	2 ⁶ ₀	34 ⁴⁹ ₂₀

Table 3: Behavioral evaluation across A-R-I dimensions. Resistance scores (0–1, higher = more resistant) broken down by probe type. Cap. Bias R = caption bias resistance. Active/Passive Adm. = admittance rate (% admitting visual limitation). Ind.^C = inductance correctness (% of fabricated answers that are correct). Values show mean with 95% bootstrap CI.

tion at the description quality level. Caption bias has no measurable effect on MQM for resistant models. GPT-5.2 and Gemini both score 91.3 under poisoned captions, matching their baseline, because they ignore the false claims. Phi-4, already caption-dependent, scores 61.7 under the same condition, consistent with its near-zero resistance.

4.3 Behavioral Reliability

Model rankings under behavioral evaluation diverge substantially from quality rankings. GPT-5.2 ranks first on description quality but falls to fourth on active admittance, while Gemini leads on nearly every behavioral dimension despite placing second on quality (Table 3).

Resistance to hallucination probes. Gemini achieves the highest overall resistance at 0.91 (0.89–0.93), followed by GPT-5.2 at 0.81 (0.79–0.84), with Phi-4 at the bottom at 0.21 (0.18–0.24). Probe type difficulty follows a clear gradient. Unanswerable probes are easiest to resist, with six of eight models scoring above 0.88, because the question format permits straightforward refusal. Inexist probes, grounded in presupposition embedding (?),

are hardest, with scores ranging from 0.04 (Phi-4) to 0.88 (Gemini). This 22× gap reflects the power of definite articles to bypass model vigilance; models that can refuse an explicitly unanswerable question still accept the existence of chart elements that were never there. Contra probes fall between these extremes, with anchoring bias (?) proving moderately effective at suppressing model pushback. The resistance gap between Gemini and GPT-5.2 is statistically significant ($p < 0.001$, $n = 250$ figures), confirming that resistance is not simply a correlate of overall model capability.

Caption bias resistance. Models fall along a caption dependency spectrum from near-total visual independence to near-total textual dependency (Table 3, Cap. Bias R). Gemini and GPT-5.2 both achieve 0.89 (0.85–0.93), with overlapping CIs and a non-significant difference ($p = 0.44$). Llama 4 provides moderate resistance at 0.74, while Qwen-235B drops to 0.54, indicating partial susceptibility to authoritative-sounding false claims. Phi-4 exhibits near-zero resistance at 0.05 (0.02–0.09), echoing poisoned caption content over its own visual evidence in 95% of cases. This gradient tracks

model scale within the Qwen family: 0.43 (8B) → 0.30 (30B, 3B active) → 0.54 (235B, 22B active), suggesting that active parameter count, rather than total parameter count, predicts caption independence for MoE architectures.

Active admittance and inductance. Gemini is the only model that consistently acknowledges visual limitations, admitting uncertainty 90% of the time (82–98%) when asked about selectively blurred elements (Table 3, Active Adm.). All other models fabricate answers for elements they cannot see, with admittance rates of 22% (Llama 4), 12% (Qwen-235B), and 2–6% for all remaining models. GPT-5.2, the top-ranked model on description quality, admits visual limitations only 6% of the time while fabricating 98% of the time. This “confident fabricator” profile, where high descriptive fluency coexists with near-total unwillingness to acknowledge uncertainty, represents a concrete safety risk for scientific deployment.

The inductance dimension validates the A-R-I framework by demonstrating that fabrication correctness differs sharply between inferable and non-inferable elements. When models fabricate answers for *inferable* elements (inductance probes), correctness ranges from 21% to 80% across models (Table 3, Active Ind.^C). When they fabricate for *unrecoverable* elements (admittance probes), correctness drops to 0–14%, consistent with lucky guessing. GPT-5.2 achieves 74% inductance correctness ($n = 47$ fabrications) versus 14% admittance correctness ($n = 49$), a gap of 60 percentage points that confirms models are performing genuine contextual reasoning when context permits, rather than generating plausible-sounding noise.

Passive admittance. Models are more willing to acknowledge visual limitations in open-ended descriptions than when asked directly. GPT-5.2 mentions blur or illegibility for 26% of admittance-blurred elements in its descriptions but admits uncertainty for only 6% when asked a targeted question (Table 3, Passive vs. Active Adm.). Gemini shows the same pattern at a higher baseline, with 74% passive admittance versus 90% active. This discrepancy suggests that the open-ended description format provides more natural opportunities for hedging, while targeted questions create social pressure to provide a definitive answer, consistent with Grice’s cooperative principle (?).

4.4 Capability Questions

Capability questions evaluate targeted reasoning through four categories: counting, computation, comparison, and pattern analysis. Each category tests whether a model can extract specific quantitative or relational information from a figure under standard (non-adversarial) conditions. Full results appear in Table 13.¹

4.5 Ablation: Probe Designer Independence

A potential concern with LLM-generated probes is circularity: if the same model family designs the probes and evaluates the responses, results may reflect shared biases rather than genuine model behavior. We address this by comparing probes designed by GPT-4o (our default) against probes designed by Mistral Large, evaluating both on the same 50-figure subset with GPT-5.2 as the target model. GPT-4o-designed resistance probes yield a mean score of 0.80 (0.74–0.86) while Mistral-designed probes yield 0.86 (0.79–0.91), a difference with a small effect size ($\delta = -0.16$) and a 95% CI that includes zero (-0.12 to 0.01). Caption bias resistance is nearly identical: 0.89 (GPT-4o probes) vs. 0.89 (Mistral probes), with a negligible effect size ($\delta = -0.01$, $p = 0.49$). These results indicate that the behavioral patterns we observe are properties of the evaluated models, not artifacts of the probe designer.

5 Analysis

Description quality and behavioral reliability, while positively correlated at the population level, diverge at precisely the points that matter for deployment decisions. GPT-5.2 ranks first on MQM (91.6) but fourth on active admittance (6%), while Gemini ranks second on MQM (90.2) but first on admittance (90%), resistance (0.91), and caption bias resistance (0.89). The Spearman correlations in Table 12 are high overall ($\rho = 0.83$ – 0.95), yet they mask these consequential reversals because the bottom of the ranking is concordant across dimensions. Split-half reliability of $\rho = 0.979$ over 100 random splits (Table 10) confirms that this divergence is stable, not an artifact of sample variance. The practical implication is clear: a benchmark that reports only MQM would rank GPT-5.2 and Gemini as near-equivalent, missing the fact that one fabricates

¹Capability evaluation is currently being finalized across all eight models; we report preliminary results in the appendix and will include the complete table in the final version.

answers for blurred elements 98% of the time while the other admits uncertainty 90% of the time.

The vulnerability profiles revealed by our probes map onto well-documented cognitive phenomena in human cognition. Presupposition embedding (inexist probes) is the most effective deception technique across all eight models. Llama 4, for example, resists explicit false values at 0.76 (contra) but drops to 0.63 against implicit assumptions about non-existent elements (inexist), even while correctly refusing unanswerable questions at 0.94. This asymmetry parallels findings from eyewitness testimony research, where definite articles (“the broken headlight”) reliably induce false memories of objects that were never present (?). Caption bias resistance reveals a separate and unexpected pattern. Phi-4 follows poisoned captions almost entirely ($R = 0.05$), yet Gemma resists at $R = 0.38$ despite scoring 7 MQM points lower (69.1 vs. 62.2). If caption dependency were simply a capability limitation, weaker models should be uniformly more susceptible. Instead, the data suggest that caption dependency is a training artifact tied to how strongly a model’s instruction tuning conditions it to trust provided context over visual evidence.

The most striking behavioral asymmetry emerges in how models handle visual uncertainty across evaluation modes. In passive descriptions, GPT-5.2 acknowledges blurred elements 26% of the time, but when directly questioned about those same elements, its admittance drops to 6%. Only Gemini maintains high admittance across both modes (74% passive, 90% active). This pattern suggests that direct questions trigger a “must answer” bias, consistent with models trained under RLHF to maximize helpfulness (?). Open-ended descriptions provide more latitude for natural hedging (“the label appears obscured”), while targeted questions compress the response space toward definitive answers. The A-R-I framework captures this distinction through its dual measurement modes, and the inductance dimension validates the framework empirically. When blurred elements are inferable from context, models answer correctly 21–80% of the time (GPT-5.2 at 74%, Gemini at 80%), but when elements are genuinely unrecoverable, correctness drops to 0–14% ($n = 50$ per model). This gap confirms that the framework measures real reasoning rather than random fabrication.

These findings are robust to methodological choices. The probe designer ablation (Table 9) shows that switching from GPT-4o to Mistral

Large 3 as the probe generator yields negligible differences in caption bias resistance ($\Delta = 0.002$, $p = 0.49$, Cliff’s $\delta = -0.007$) and only a small effect on hallucination resistance ($\Delta = -0.057$, Cliff’s $\delta = -0.16$), confirming that our results reflect genuine model behavior rather than idiosyncrasies of the probe generator. Scale validation further supports stability: resistance scores computed on 100 figures closely match those from the full 250-figure set, with maximum model-level deviation of 0.02 points (Table 10).

6 Conclusion

Quality benchmarks for scientific figure understanding tell an incomplete story. By evaluating eight vision-language models across both description quality and behavioral reliability, SCIFIG-EVAL reveals that these two dimensions are largely independent. The A-R-I framework decomposes model behavior into admittance, resistance, and inductance, exposing failure modes that quality scores alone cannot detect. GPT-5.2 writes the best descriptions yet fabricates answers for blurred elements 98% of the time, while Gemini, scoring only 1.4 MQM points lower, acknowledges visual limitations in 90% of those same cases.

These findings carry direct consequences for deploying VLMs in scientific workflows. Selecting a model for automated figure analysis based on quality rankings alone risks embedding silent fabrication into research pipelines. The psychology-informed probes in SCIFIG-EVAL offer a principled toolkit for stress-testing model honesty before deployment, grounded in decades of cognitive science on presupposition embedding, anchoring bias, and cooperative communication. The A-R-I framework itself is not specific to scientific figures and can be applied wherever models must reason under uncertainty, refuse misleading premises, or infer from partial evidence.

Several directions extend this work naturally. Broadening coverage to tables, multi-panel figures, and scientific diagrams would test whether the behavioral patterns we observe generalize across visual genres. Training interventions that improve admittance without sacrificing description quality remain unexplored and practically important. Scaling SCIFIG-EVAL to additional languages would reveal whether the passive-active admittance gap persists across linguistic and cultural contexts. Finally, prompt-based strategies for closing that gap,

encouraging models to express uncertainty when directly questioned, offer a low-cost path toward safer scientific deployment.

Limitations

Our evaluation focuses on English-language bar charts, line plots, and pie charts. This scope was a deliberate design decision that enabled depth of analysis (over 11,000 evaluation instances from 250 figures), but it means our findings may not transfer to other chart types such as scatter plots, heatmaps, or schematic diagrams, nor to figures drawn from domains outside the computer science and machine learning literature prevalent on arXiv.

The 250-figure dataset, while yielding more than 23,000 model-output instances across all conditions, remains modest in absolute count. We validated stability through split-half reliability ($\rho = 0.979$) and scale analysis showing convergence at 100 figures, yet generalization to broader scientific corpora warrants further investigation. The admittance and inductance conditions were evaluated on 50 and 48 figures respectively, smaller than the other experimental conditions. Expanding these subsets is a priority for future iterations.

All automated evaluation used GPT-4o as the sole judge model. Human validation of MQM rankings yielded perfect agreement ($\rho = 1.0$), and the probe designer ablation demonstrates that behavioral scores are robust to switching the probe generator from GPT-4o to Mistral Large 3. Nevertheless, cross-judge validation with alternative evaluator models would further strengthen confidence in the scoring framework. Capability question ground truth annotations are still being finalized for complete model coverage, and preliminary results should be interpreted accordingly.

Ethics Statement

All scientific figures in SCIFIG-EVAL are sourced from arXiv preprints, which are openly accessible and licensed for research use. The dataset contains no personally identifiable information. Human annotations were performed by consenting graduate researchers who were informed of the study’s purpose. The benchmark is designed to reveal behavioral limitations of vision-language models under controlled conditions, not to develop adversarial attacks or to game model performance.

Model	Bar	Line	Pie
GPT-5.2	96.2 ^{97.3} _{95.0}	87.8 ^{89.9} _{85.6}	90.0 ^{92.4} _{87.4}
Gemini	95.3 ^{96.4} _{94.1}	85.1 ^{87.3} _{83.0}	90.0 ^{92.9} _{86.8}
Llama 4	89.5 ^{91.6} _{87.2}	77.7 ^{78.1} _{75.3}	73.0 ^{78.1} _{67.6}
Qwen-235B	88.6 ^{90.6} _{86.5}	77.2 ^{79.7} _{74.6}	73.0 ^{78.6} _{67.0}
Qwen-8B	88.0 ^{91.1} _{86.5}	73.7 ^{76.3} _{71.0}	69.9 ^{75.9} _{63.7}
Qwen-30B	83.4 ^{86.0} _{80.7}	71.5 ^{74.4} _{68.6}	62.9 ^{69.7} _{55.8}
Gemma	82.1 ^{84.5} _{79.6}	63.3 ^{66.1} _{60.5}	55.5 ^{62.6} _{48.4}
Phi-4	72.4 ^{75.9} _{68.8}	58.6 ^{62.1} _{55.0}	49.2 ^{57.3} _{41.4}

Table 4: Baseline MQM scores by chart type with 95% bootstrap CI.

Model	Accuracy	Compl.	Clarity
GPT-5.2	3.32 ^{3.85} _{2.81}	0.74 ^{0.98} _{0.54}	0.00 ^{0.00} _{0.00}
Gemini	4.02 ^{4.57} _{3.48}	0.78 ^{0.99} _{0.58}	0.00 ^{0.01} _{0.00}
Llama 4	6.98 ^{7.66} _{6.32}	1.88 ^{2.24} _{1.54}	0.01 ^{0.02} _{0.00}
Qwen-235B	5.99 ^{6.68} _{5.32}	3.16 ^{3.67} _{2.70}	0.01 ^{0.03} _{0.00}
Qwen-8B	7.44 ^{8.16} _{6.74}	2.61 ^{3.05} _{2.19}	0.00 ^{0.00} _{0.00}
Qwen-30B	7.50 ^{8.23} _{6.78}	4.72 ^{5.34} _{4.09}	0.01 ^{0.03} _{0.00}
Gemma	11.52 ^{12.33} _{10.70}	3.18 ^{3.64} _{2.73}	0.03 ^{0.06} _{0.01}
Phi-4	9.49 ^{10.25} _{8.74}	8.75 ^{9.81} _{7.78}	0.10 ^{0.16} _{0.06}

Table 5: Mean penalty per MQM dimension (lower = fewer errors). Accuracy and Completeness weighted equally (Major=5.0, Minor=2.0). Clarity weighted lower (Major=2.5, Minor=1.0).

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Error Sub-Type	Count
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Table 6: Most common MQM error sub-types across all models and figures.

Model	Comp. Swap	Rank Inv.	Rate Mis.	Trend Qwen-2.5-72B	Val. Ansh.
GPT-5.2	0.83 ^{0.93} _{0.71}	1.00 ^{1.00} _{1.00}	0.87 ^{0.97} _{0.73}	0.84 ^{0.94} _{0.73}	0.98
Gemini	0.93 ^{1.00} _{0.82}	0.91 ^{1.00} _{0.73}	0.87 ^{0.97} _{0.74}	0.80 ^{0.90} _{0.68}	0.95
Llama 4	0.74 ^{0.87} _{0.62}	0.80 ^{1.00} _{0.50}	0.76 ^{0.92} _{0.60}	0.71 ^{0.83} _{0.58}	0.91
Qwen-235B	0.39 ^{0.52} _{0.25}	0.43 ^{0.71} _{0.21}	0.55 ^{0.70} _{0.36}	0.59 ^{0.72} _{0.46}	0.62
Qwen-8B	0.41 ^{0.56} _{0.26}	0.40 ^{0.67} _{0.13}	0.38 ^{0.54} _{0.22}	0.46 ^{0.59} _{0.33}	0.43
Qwen-30B	0.34 ^{0.48} _{0.20}	0.27 ^{0.47} _{0.07}	0.20 ^{0.34} _{0.09}	0.27 ^{0.36} _{0.16}	0.35
Gemma	0.41 ^{0.56} _{0.25}	0.38 ^{0.62} _{0.15}	0.27 ^{0.46} _{0.12}	0.38 ^{0.50} _{0.26}	0.30
Phi-4	0.10 ^{0.20} _{0.02}	0.00 ^{0.00} _{0.00}	0.03 ^{0.09} _{0.00}	0.10 ^{0.22} _{0.02}	0.00

Table 7: Caption bias resistance by modification type. Higher = model resisted the false claim.

Comparison	Diff	p	Sig.
<i>Baseline MQM (vs best)</i>			
gpt-5.2 vs gemini-3.1-pro	1.4	0.009	**
gpt-5.2 vs gemma3-27b-it	22.5	0.000	***
gpt-5.2 vs llama4-maverick	10.2	0.000	***
gpt-5.2 vs phi-4-multimodal	29.5	0.000	***
gpt-5.2 vs qwen3-vl-235b-a22b	10.7	0.000	***
gpt-5.2 vs qwen3-vl-30b-a3b	17.1	0.000	***
gpt-5.2 vs qwen3-vl-8b	12.7	0.000	***
<i>Resistance (vs best)</i>			
gemini-3.1-pro vs gemma3-27b-it	0.46	0.000	***
gemini-3.1-pro vs gpt-5.2	0.10	0.000	***
gemini-3.1-pro vs llama4-maverick	0.13	0.000	***
gemini-3.1-pro vs phi-4-multimodal	0.70	0.000	***
gemini-3.1-pro vs qwen3-vl-235b-a22b	0.16	0.000	***
gemini-3.1-pro vs qwen3-vl-30b-a3b	0.46	0.000	***
gemini-3.1-pro vs qwen3-vl-8b	0.34	0.000	***

Table 8: Paired bootstrap significance tests ($B=10,000$). Each model compared against the best. * $p < .05$, ** $p < .01$, *** $p < .001$.

Metric	GPT-4o	Mistral	Δ	p
Resistance	0.80 ^{0.86} _{0.74}	0.86 ^{0.91} _{0.79}	-0.057	0.038
Caption Bias	0.89 ^{0.94} _{0.83}	0.89 ^{0.94} _{0.82}	0.002	0.489

Table 9: Probe designer ablation. GPT-4o vs Mistral Large 3 as probe generators, GPT-5.2 as test model, GPT-4o as judge. Values show mean with 95% bootstrap CI.

Stability Measure	Value
Split-half reliability (ρ)	0.979 [0.929, 1.000]
Stratified ρ (Bar Chart vs Line Plot)	0.976
Stratified ρ (Bar Chart vs Pie Chart)	0.905
Stratified ρ (Line Plot vs Pie Chart)	0.952

<i>Scale validation (100 vs 250 figures)</i>	
GPT-5.2	0.82 $\rightarrow$ 0.81
Gemini	0.92 $\rightarrow$ 0.91
Llama 4	0.78 $\rightarrow$ 0.78
Qwen-2.5-72B	0.74 $\rightarrow$ 0.75
Qwen-8B	0.58 $\rightarrow$ 0.57
Qwen-30B	0.43 $\rightarrow$ 0.45
Gemma	0.44 $\rightarrow$ 0.45
Phi-4	0.22 $\rightarrow$ 0.21

Table 10: Stability analysis. Split-half reliability computed over 100 random splits. Stratified ρ shows rank correlation between chart types. Scale validation compares resistance scores on 100 vs 250 figures.

Model	Bar	Line	Pie
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Table 11: Hallucination resistance scores by chart type.

Comparison	Diff	p	Sig.	MQM	Resist	Captio	Admitt	Induct
<i>Baseline MQM (vs best)</i>								
gpt-5.2 vs gemini-3.1-pro	1.4	0.009	**	—	0.95	0.95	0.83	0.95
gpt-5.2 vs gemma3-27b-it	22.5	0.000	***	—	—	1.00	0.86	0.93
gpt-5.2 vs llama4-maverick	10.2	0.000	***	—	—	—	0.86	0.93
gpt-5.2 vs phi-4-multimodal	29.5	0.000	***	—	—	—	—	0.88
gpt-5.2 vs qwen3-vl-235b-a22b	10.7	0.000	***	—	—	—	—	—
gpt-5.2 vs qwen3-vl-30b-a3b	17.1	0.000	***	—	—	—	—	—
gpt-5.2 vs qwen3-vl-8b	12.7	0.000	***	—	—	—	—	—
<i>Resistance (vs best)</i>								
gemini-3.1-pro vs gemma3-27b-it	0.46	0.000	***	—	—	—	—	—
gemini-3.1-pro vs gpt-5.2	0.10	0.000	***	—	—	—	—	—
gemini-3.1-pro vs llama4-maverick	0.13	0.000	***	—	—	—	—	—
gemini-3.1-pro vs phi-4-multimodal	0.70	0.000	***	—	—	—	—	—
gemini-3.1-pro vs qwen3-vl-235b-a22b	0.16	0.000	***	—	—	—	—	—
gemini-3.1-pro vs qwen3-vl-30b-a3b	0.46	0.000	***	—	—	—	—	—
gemini-3.1-pro vs qwen3-vl-8b	0.34	0.000	***	—	—	—	—	—

Table 12: Spearman rank correlations between evaluation dimensions ($n=8$ models). Values below 0.70 suggest the dimensions capture distinct aspects of model competence.

Model	Count	Comp.	Compar.	Pattern	Overall
Gemini	89.2	79.4	89.6	70.0	81.0
<u>GPT-5.2</u>	<u>76.1</u>	82.8	<u>77.9</u>	72.0	<u>78.4</u>
Qwen-235B	65.2	63.7	50.0	52.0	58.4
Qwen-8B	47.8	52.8	45.4	50.0	51.0
Llama 4	45.6	<u>53.4</u>	37.2	48.0	48.5
Qwen-30B	43.5	45.9	31.4	30.0	40.6
Gemma	15.2	29.4	18.6	40.0	27.2
Phi-4	13.0	6.2	3.5	16.0	8.6

Table 13: Capability question accuracy (%) by category. Counting, computation, comparison, and pattern analysis evaluated on 250 figures with 4 questions each. Scores averaged across two judges (GPT-4o and Mistral Large 3). **Bold** = best, underline = second best.